

# Zuhayr Huseni

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## EDUCATION

### Georgia Institute of Technology

Master of Science in Computer Science (M.S.), Machine Learning Specialization

Atlanta, GA (Online)

Incoming Fall 2026 - Expected 2028

### University of Connecticut

Bachelor of Science in Computer Science (B.S.)

Storrs, CT

May 2025

Cumulative GPA: 3.76

**Relevant Coursework:** Computer Architecture, Systems Programming, Algorithms and Complexity, Artificial Intelligence, Big Data Analytics, Cyber Security, Principles of Databases, Software Engineering

**Achievements:** Honors Scholar and Laureate, STEM Scholar, Dean's List Engineering, Gloshoc Hackathon Winner

## EXPERIENCE

### Software Engineer

July 2025 - Present

PayPal

San Jose, CA

- Design and maintain distributed Spring Boot microservices on GCP that process transaction, account, and payment data for PayPal's global Suspicious Activity Report platform, implementing encrypted data pipelines and secure service-to-service authentication that protect customer banking and card details while satisfying jurisdiction-specific regulatory requirements.
- Built a company-wide test-data CLI on Claude Code that provisions custom test accounts with synthetic bank accounts, credit cards, and identity attributes on demand, replacing manual cross-team setup requests and giving engineers realistic end-to-end scenarios in minutes.
- Developed an internal agent harness integrating Claude Desktop and Claude Code with internal MCP servers, handling tokenization and authentication against protected engineering services; adopted org-wide for debugging, code analysis, and workflow automation.
- Expanded unit and integration test coverage and introduced WireMock dependency simulation to eliminate flaky external calls, improving CI/CD pipeline reliability and shortening deployment feedback cycles.
- Investigate production incidents across distributed compliance services using Datadog, Splunk, Oracle SQL, and structured logs, resolving transaction failures, data inconsistencies, and authentication regressions during major platform upgrades.

### Undergraduate Researcher - 5G Systems and Performance

April 2024 - September 2024

University of Connecticut

Storrs, CT (Remote)

- Benchmarked 5G throughput and latency across four open-air interfaces on Linux, scripting traffic generation, node simulation, and system-log analysis in Python and shell to isolate stability regressions in real time.
- Implemented tunneling between gNodeB and UE nodes and built C-based real-time latency and throughput visualization with the srsGUI library to identify bottlenecks and optimize network performance.

### Full Stack Software Engineer Intern

June 2024 - August 2024

Forms + Surfaces

Pittsburgh, PA

- Shipped a React and Python/Flask application translating natural-language queries into SQL against a Made2Manage ERP database via the OpenAI API, cutting data-retrieval time by 40% and reducing manual reporting effort.
- Automated product design validation with Python scripting against the SolidWorks API and PyFEM finite-element simulations to model thermal and environmental stress conditions, accelerating production timelines by 50% and cutting roughly \$10,000 in annual physical prototyping costs.

### AI Development Engineer Intern

February 2024 - April 2024

Propel Flow

Hood River, OR (Remote)

- Built and deployed Python/TensorFlow inference services on AWS EC2 with Lambda-based serverless processing, delivering client-facing ML capabilities at an estimated \$40,000 in annual client savings.
- Tuned GPT-4 based recommendation pipelines and implemented MongoDB/PyMongo indexing strategies to improve retrieval latency and accuracy for financial insights.

## PROJECTS

### OneStopShopHousing.AI

September 2024 - May 2025

- Built scalable web scrapers consolidating housing listings from multiple third-party platforms into a normalized SQLite database, with an OpenAI-backed layer translating natural-language queries into optimized SQL.
- Developed the full-stack React and Node.js/Express application with secure authentication, comparison and favorites features, commute-time visualization, and modular component architecture.

## TECHNICAL SKILLS

**Languages:** Python, Java, C/C++, SQL, JavaScript, Bash

**Cloud & Infrastructure:** GCP, AWS (EC2, Lambda), Kubernetes, Docker, Linux, CI/CD

**Distributed Systems & Backend:** Spring Boot, REST Microservices, Flask, Node.js/Express, React

**Observability & Testing:** Datadog, Splunk, Structured Logging, JUnit, WireMock, Git

**Data & ML Tooling:** Oracle SQL, MongoDB, DynamoDB, SQLite, TensorFlow, OpenAI API, MCP Servers, Claude Code